Wireless Internet Of Things Lab

Versuchs-Nr.	Course Exercise 1
Versuchsname	EM Simulation of Antenna Characteristics
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Betreuender Dozent	Prof. Dr.-Ing. Jens Werner

Task

There are 2 parts in this exercise that need to be completed. Both tasks require extensive knowledge using the FEKO simulations program, especially CADFeko and POSTFeko. Using these 2 applications, we must complete both the first and second task according per instructions and finally at the end compare values from theoretical and simulated calculations.

Task 1

Firstly, we must create a wire dipole antenna in FEKO targeting the centre frequency from the table. As my matrix number ends in 1, my data is as follows.

- Center Frequency in MHz: 120
- Theoretical length in m: 1.25

Wire radius is then set to an appropriate measure as stated in the task, radius = 5mm. The frequency is also set in a sweep of +/- 25% from our center frequency which is 90-150 MHz. After creating the Wire, a voltage source is inputted directly in the middle of the wire using the default settings.

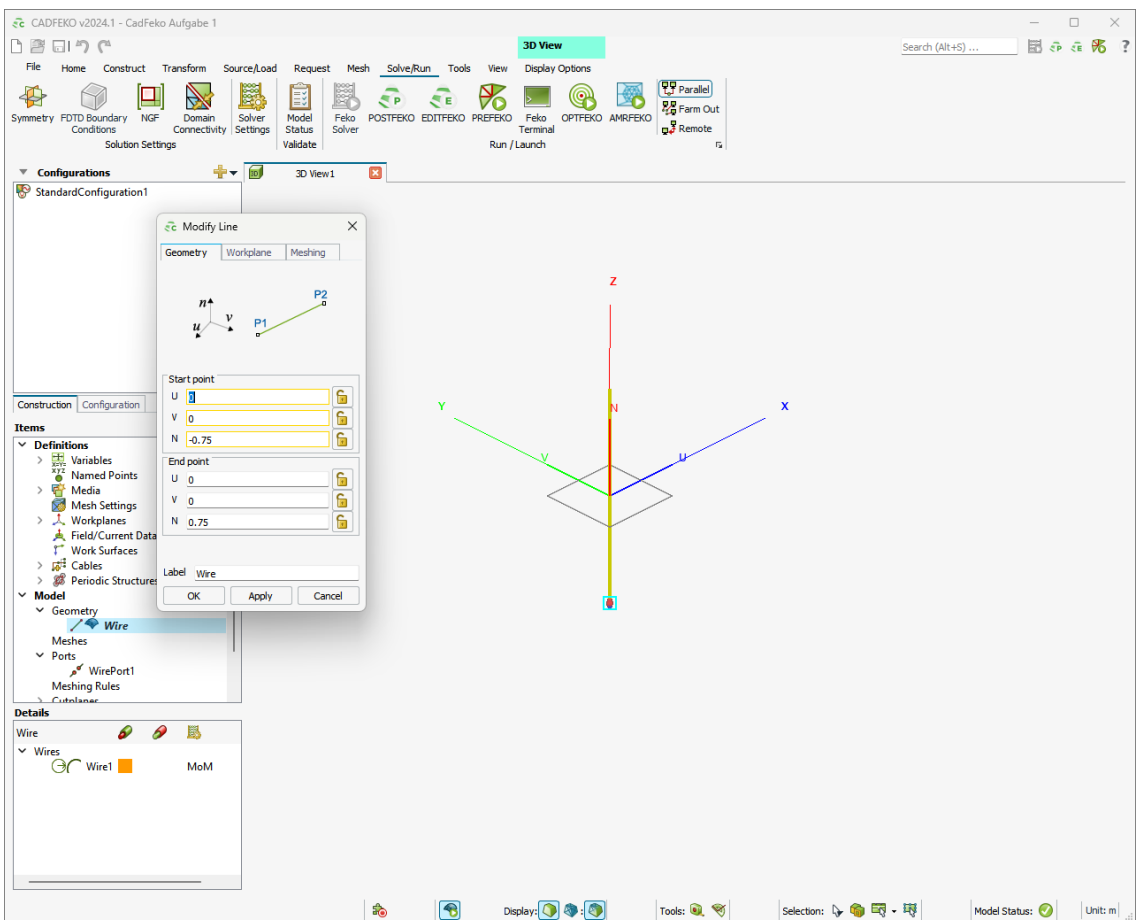

Figure 1: Dipole Wire

A far field gain is also inserted to further enhance the simulation.

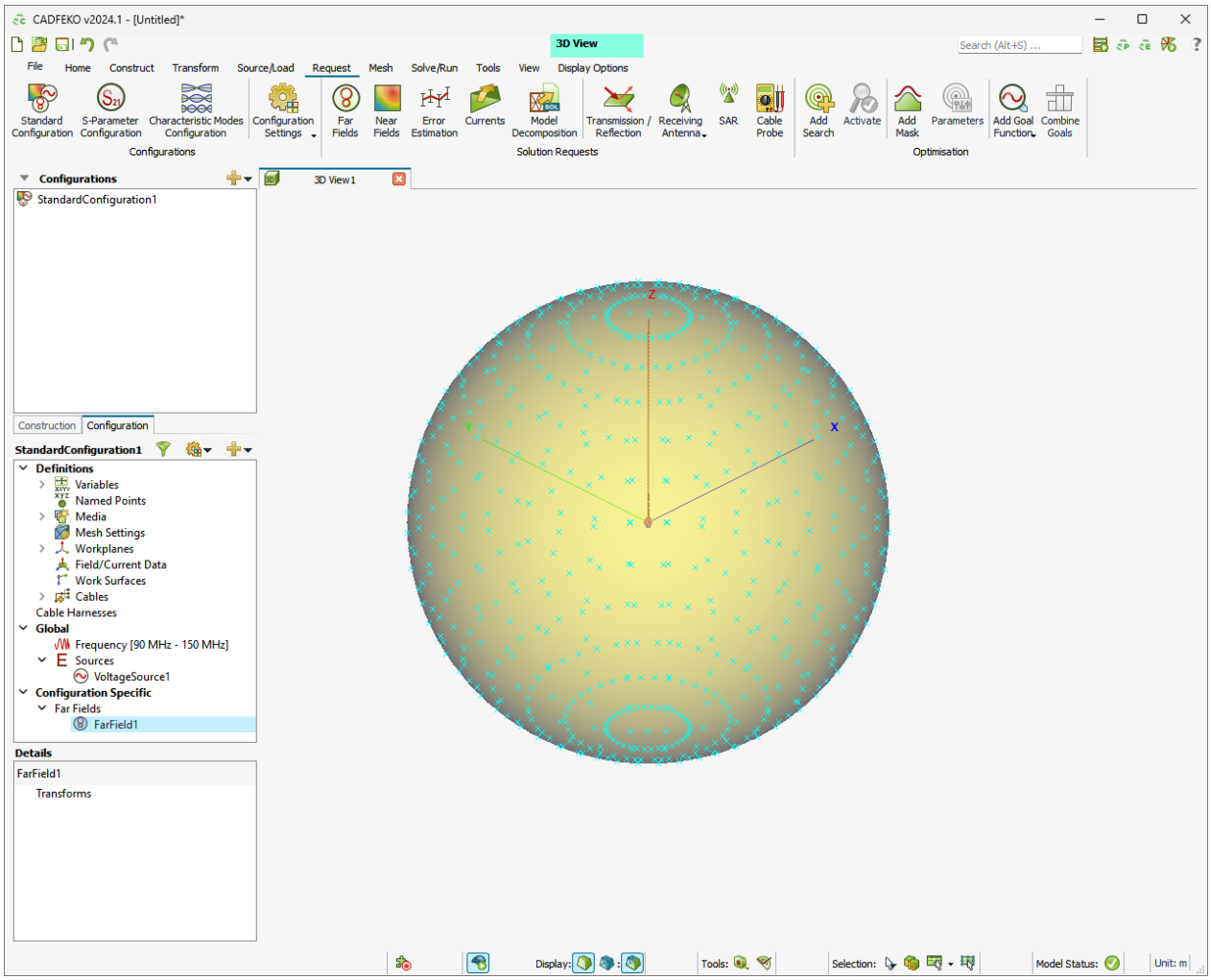

Figure 2: Far Field Gain

Next, we plot a graph using POSTFEKO in order to obtain the values required for this experiment.

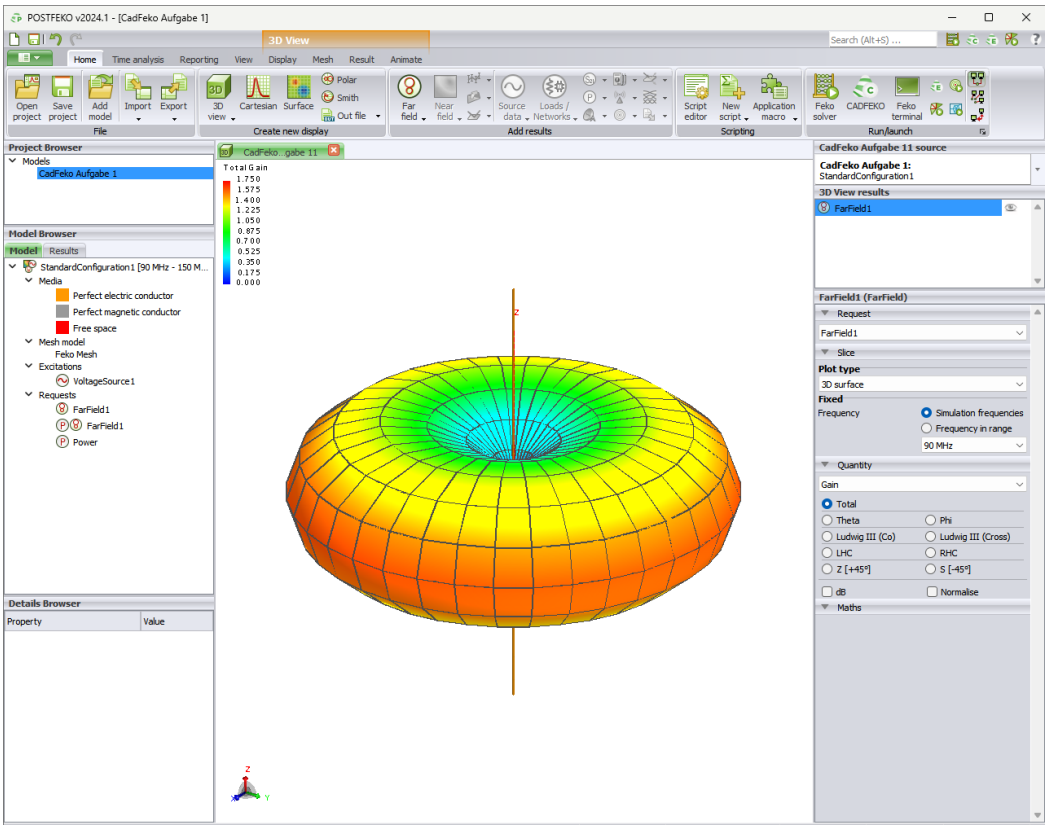

Figure 3: Far Field when frequency is 90 MHz

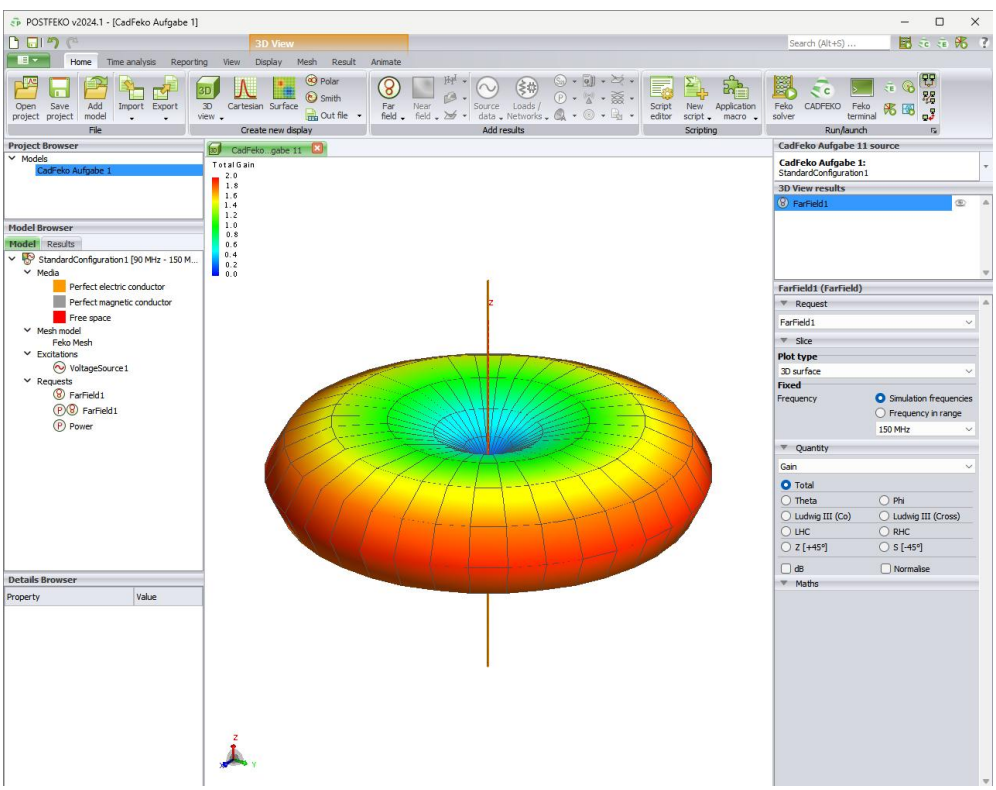

Figure 4: Far Field when frequency is 150 MHz

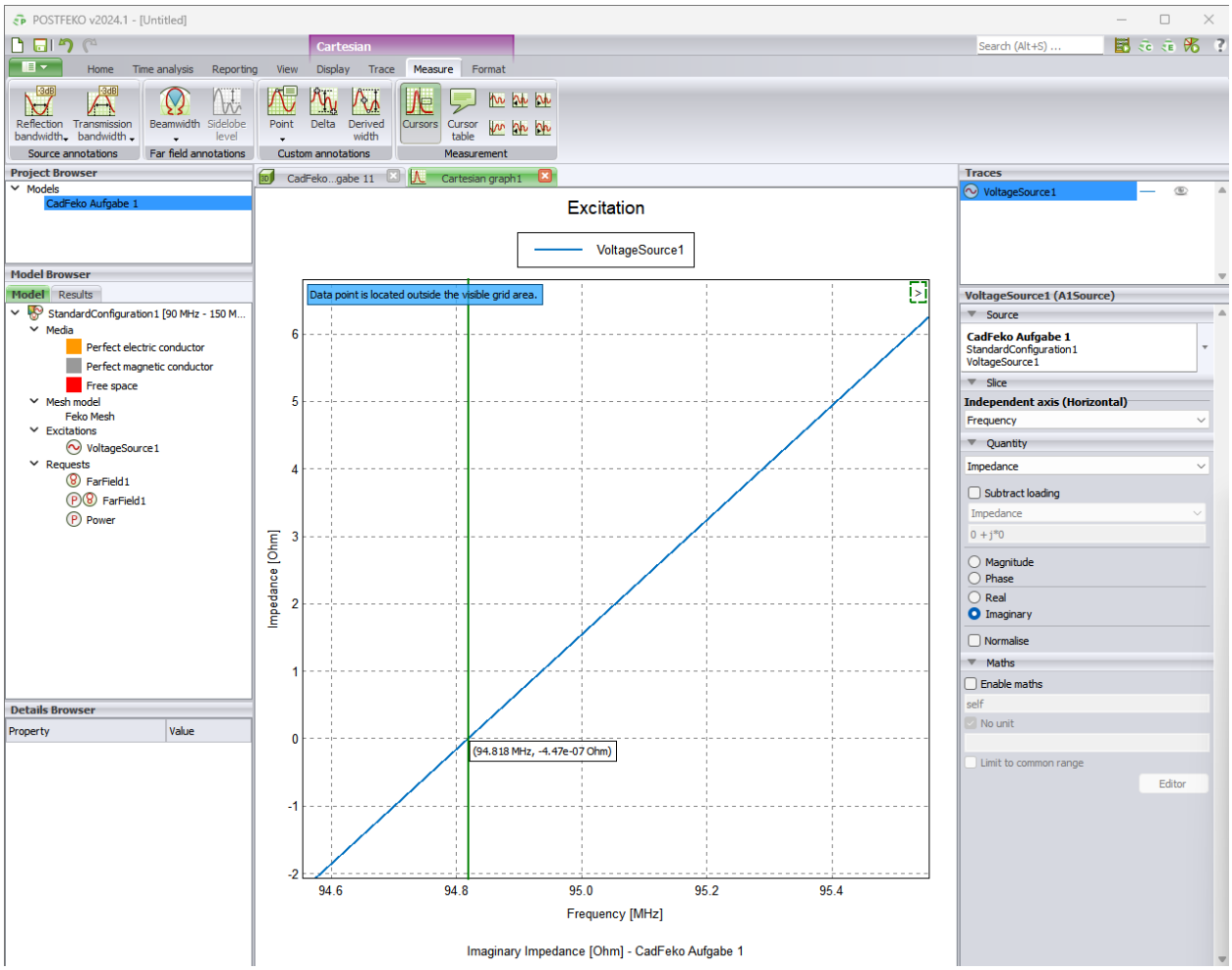

Figure 5: Imaginary Graph Impedance against frequency

We arrange a cursor, so that we can plot the point when the imaginary impedance is closest to zero and record the frequency at that point. Afterwards, the real frequency is used to determine the real impedance and reflection coefficient when the imaginary impedance is close to 0.

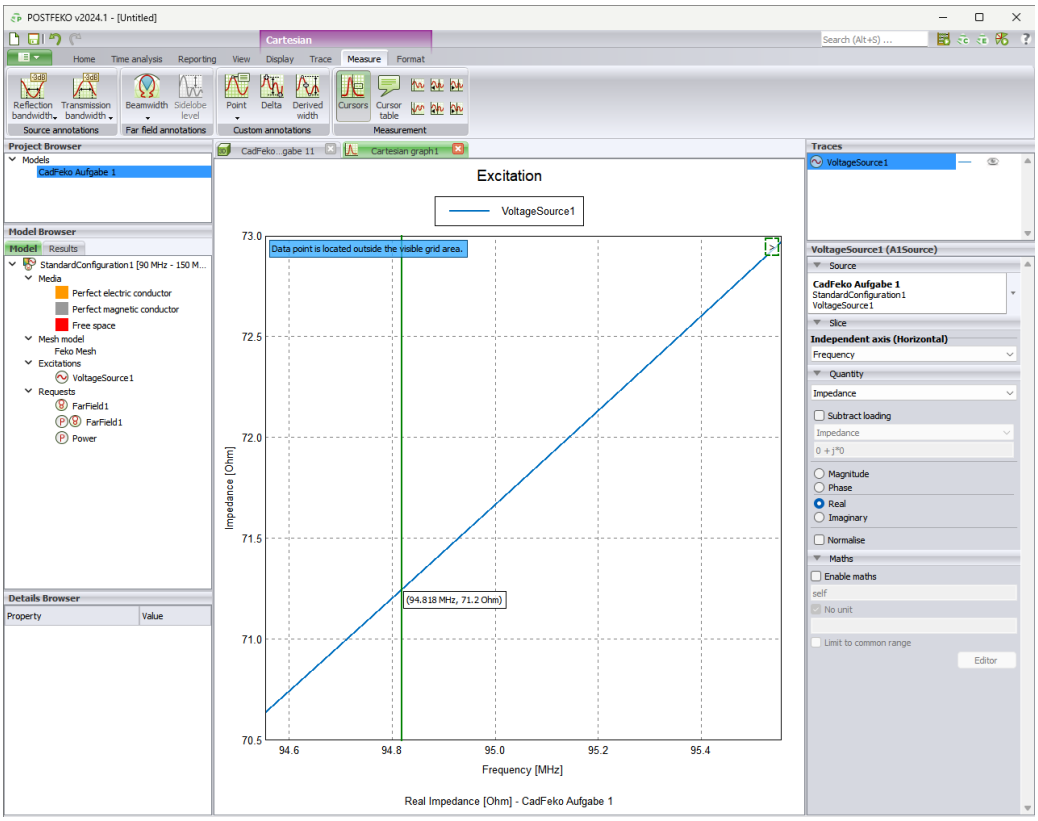

Figure *6: Real Impedance when imaginary Impedance is 0*

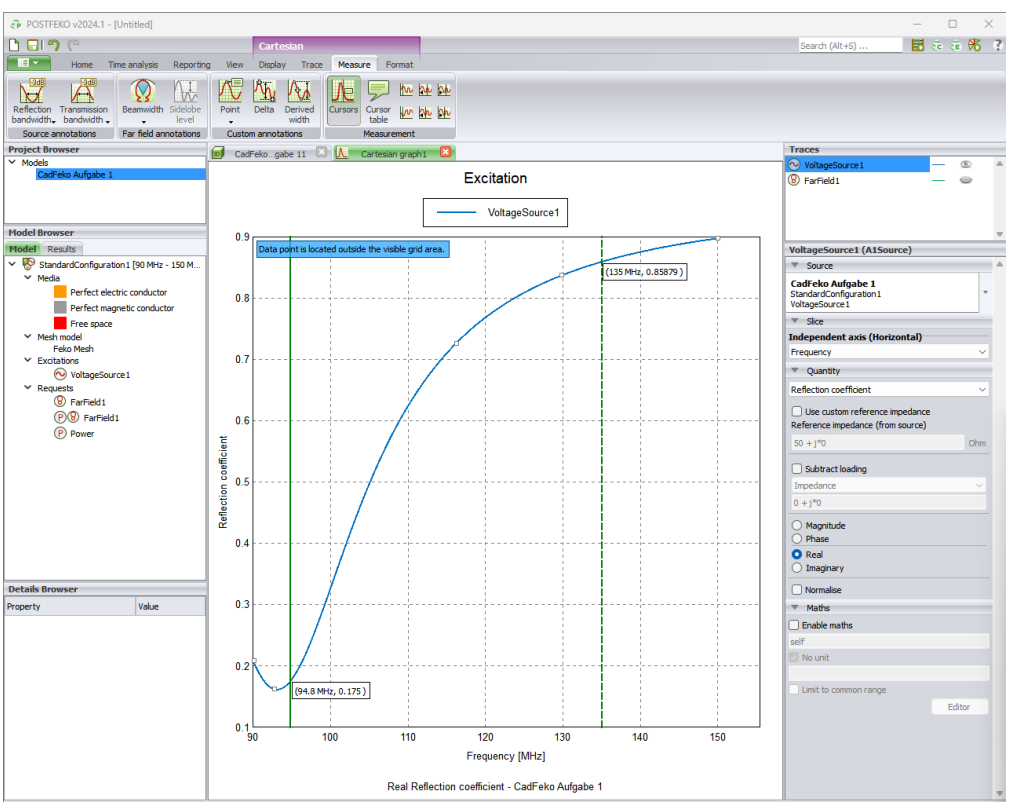

Figure *7: Real Reflection Coefficient when imaginary Impedance is 0*

Now, we use that very same Impedance in our Voltage Source and observe the changes. We also need to recalibrate our cursor so that the imaginary impedance is again close to 0.

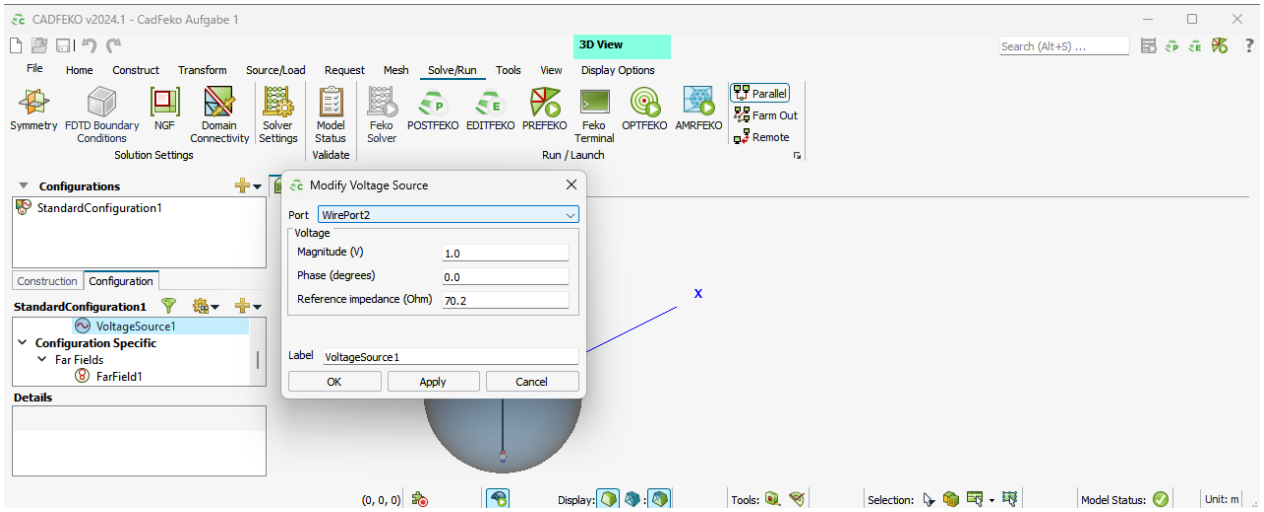

Figure *8: Voltage Source with new Impedance*

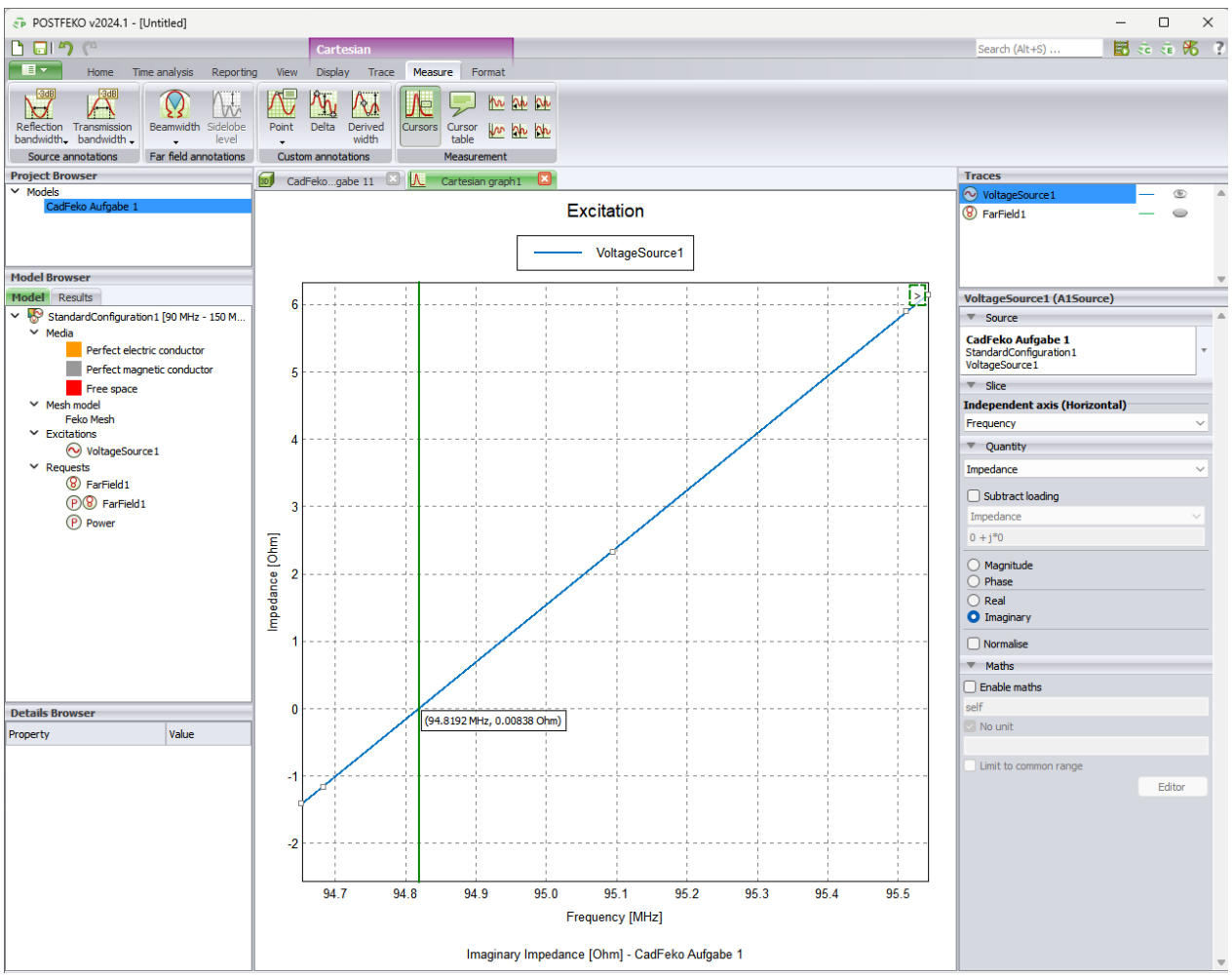

Figure *9: new imaginary Impedance = 0*

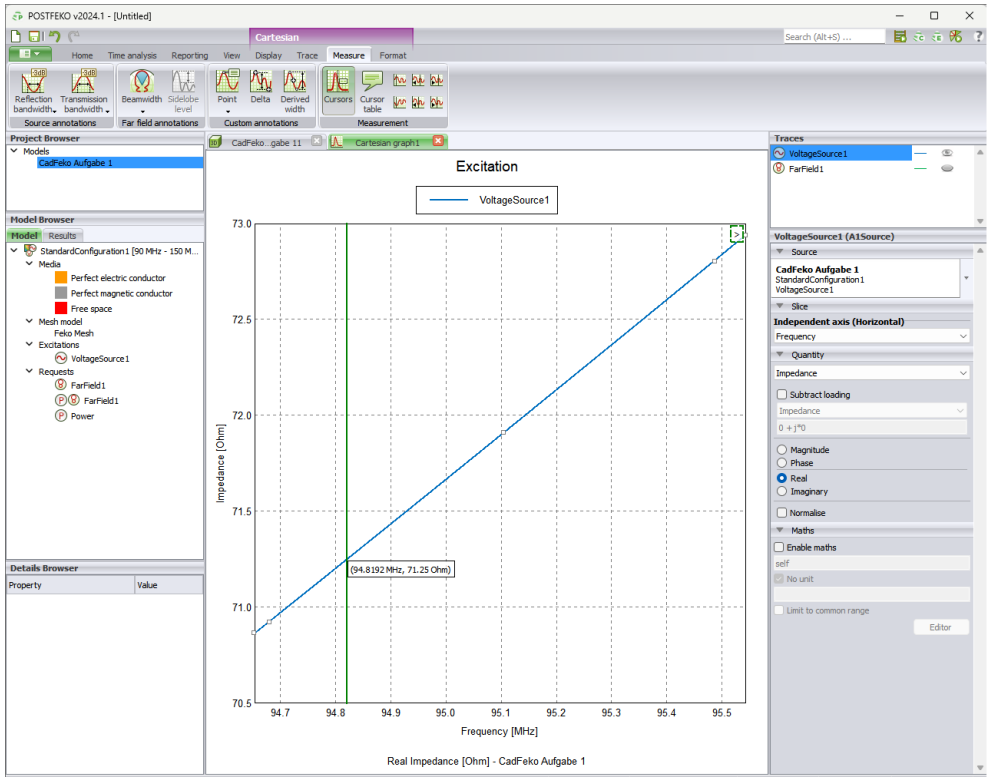

Figure *10: new Real Impedance*

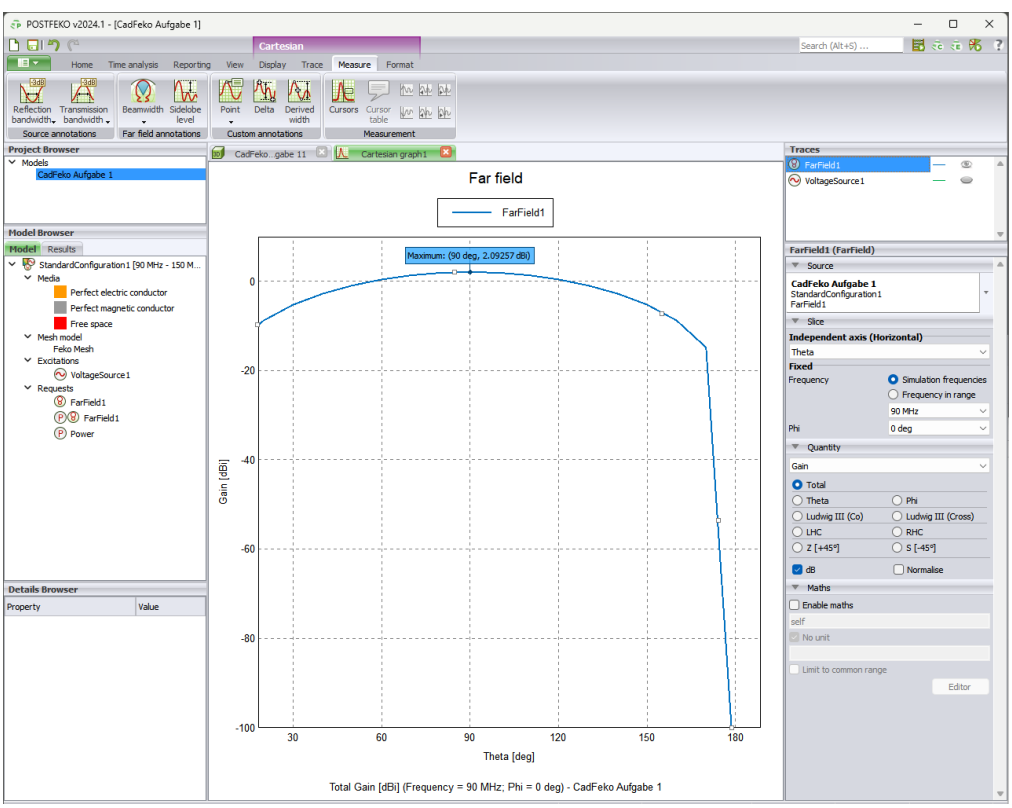

Figure *11: Antenna Gain in dB against Theta*

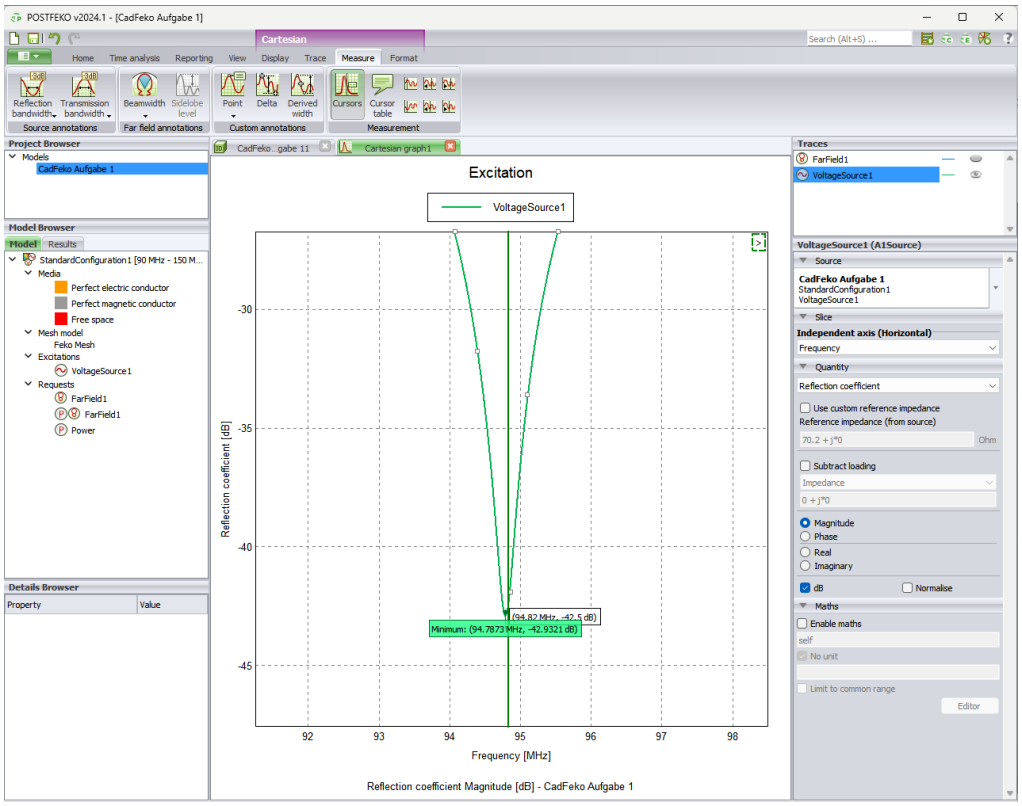

Figure *12: Reflection Coefficient in dB against frequency*

From these simulations, we can conclude that our center and optimal frequency should be around 94.82 MHz, maximal antenna Gain is 2.09257 dBi and impedance is 71,2 Ohm. These values can be recorded to be used in future calculations.

Task 2

For Task 2, we need to first create another antenna and setup a variable distance for distances of 2 m, 5 m, 10 m, 20 m, 50 m, 100 m, 200 m, 500 m, 1000 m and 2000 m.

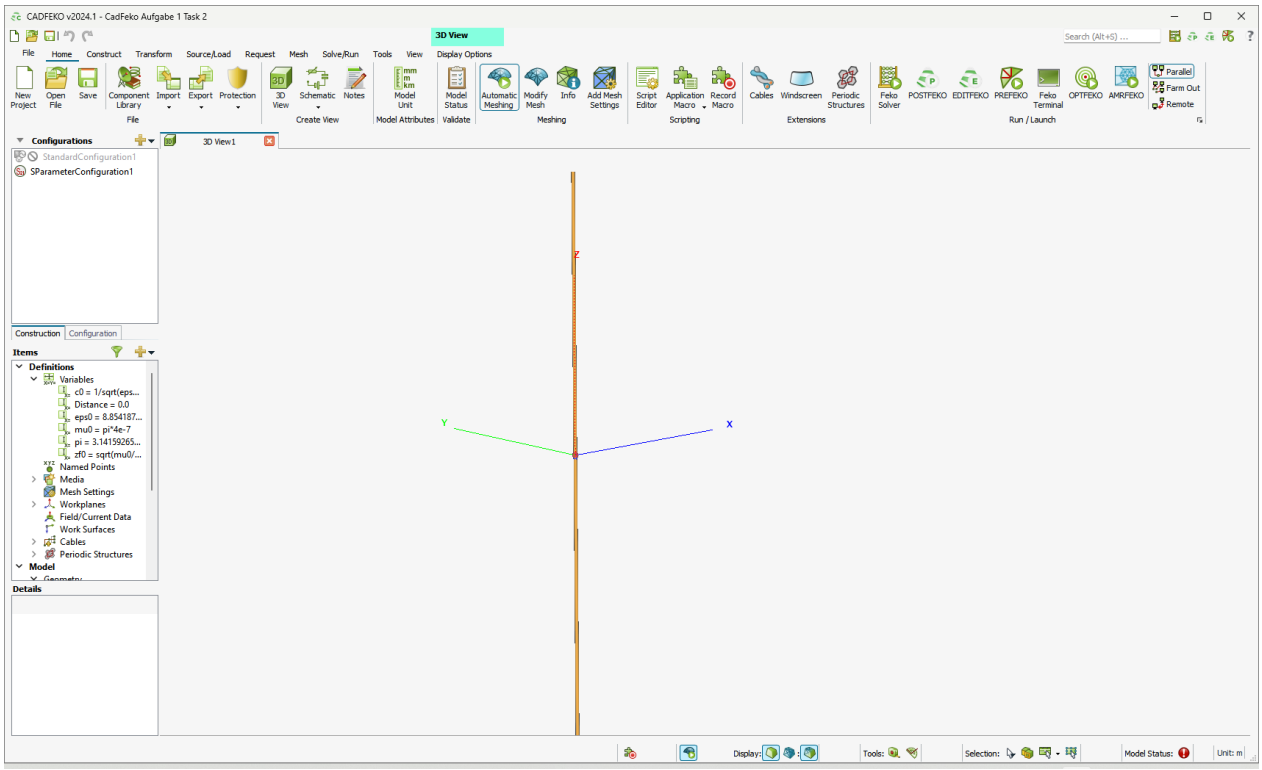

Figure *13: 2 Dipole wires with distance as variable between each other*

We also need to set up another voltage source in the middle of the other wire with the same impedance as in task 1. From that point, we need to create another configuration suitable for S parameters containing both wires with 71.2 impedance.

Then, we disable the previous configuration. Finally, we go over to the Application macro in CADFEKO to run a parameter sweep with all our distance values as variable.

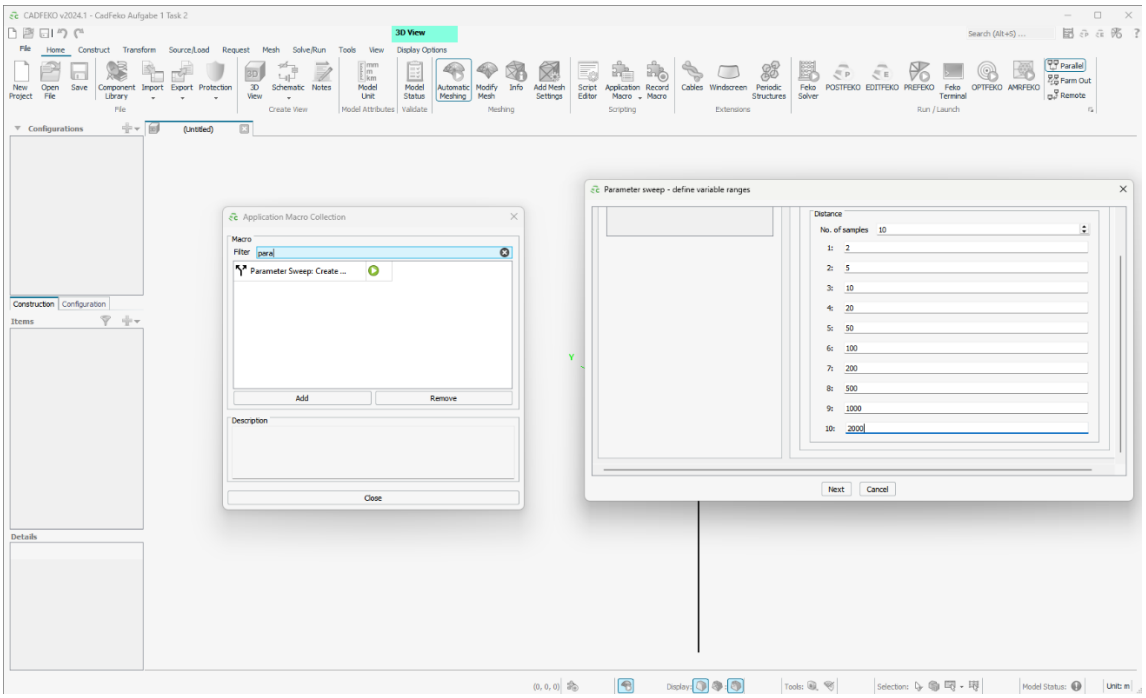

Figure *14: Parameter Sweep*

After running the parameter sweep, we open POSTFEKO and open all files that were created after the parameter sweep. Each configuration is then plotted onto a graph using S2,1 Parameter and values are collected for comparison.

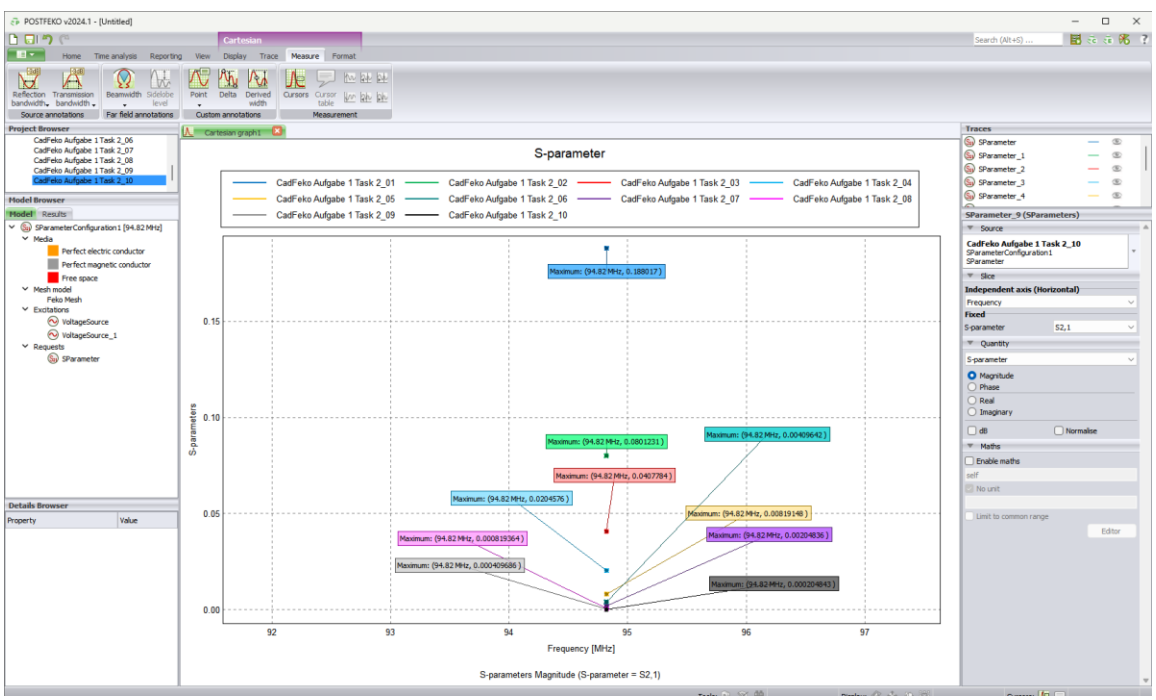

Figure *15: S Parameter graph*

The values are inputted into a table and plotted into a graph.

Table 1: S Parameter

distance in m	FSPL in S Parameter	FSPL in dB
2	0.188017	-14.516058
5	0.0801231	-21.924845
10	0.0407784	-27.791396
20	0.0204576	-33.782906
50	0.00819148	-41.732752
100	0.00409642	-47.75191
200	0.00204836	-53.771874
500	0.000819364	-61.730462
1000	0.000409686	-67.750978
2000	0.000204843	-73.771577

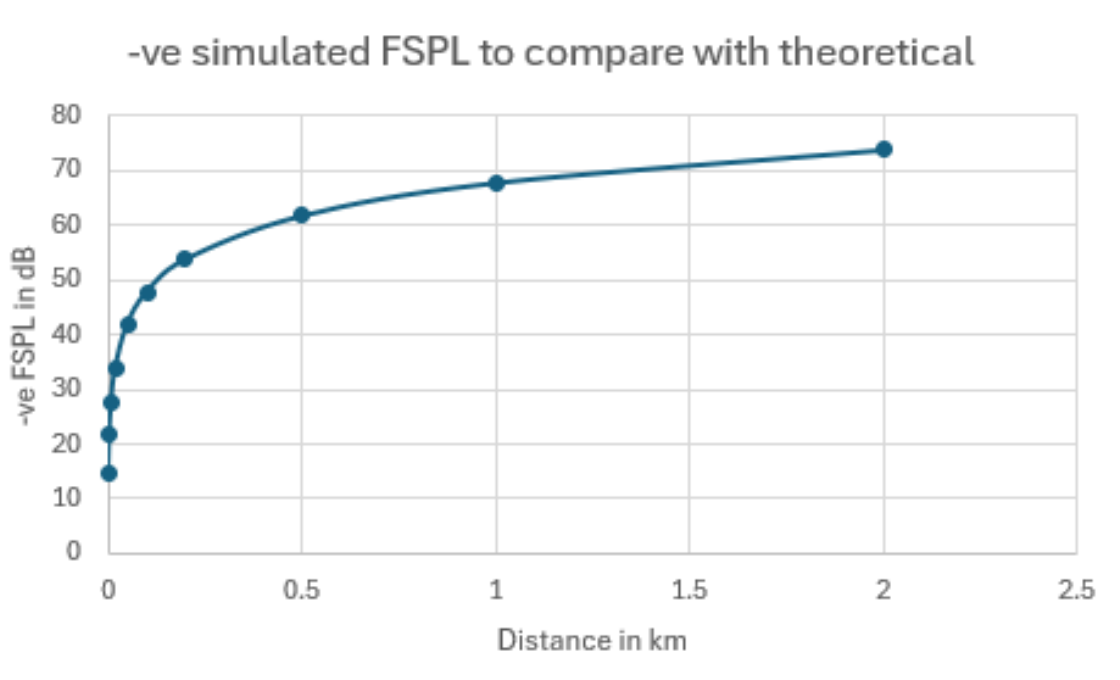

Figure *16: simulated FSPL graph*

The graph was plotted with negative values to make a closer comparison with theoretical values.

The values from Task 1 were also used to calculate the corresponding theoretical FSPL for this experiment using the formula:

$$FSPL\ in\ dB = 20log_{10}(d)\ +\ 20log_{10}(f)\ + 32.44$$

The results are inserted into a table and plotted into a graph.

Table 1: S Parameter

distance in m	distance in km	frequency in MHz	FSPL in dB
2	0.002	94.82	17.9985989
5	0.005	94.82	25.9573991
10	0.01	94.82	31.977999
20	0.02	94.82	37.9985989
50	0.05	94.82	45.9573991
100	0.1	94.82	51.977999
200	0.2	94.82	57.9985989
500	0.5	94.82	65.9573991
1000	1	94.82	71.977999
2000	2	94.82	77.9985989

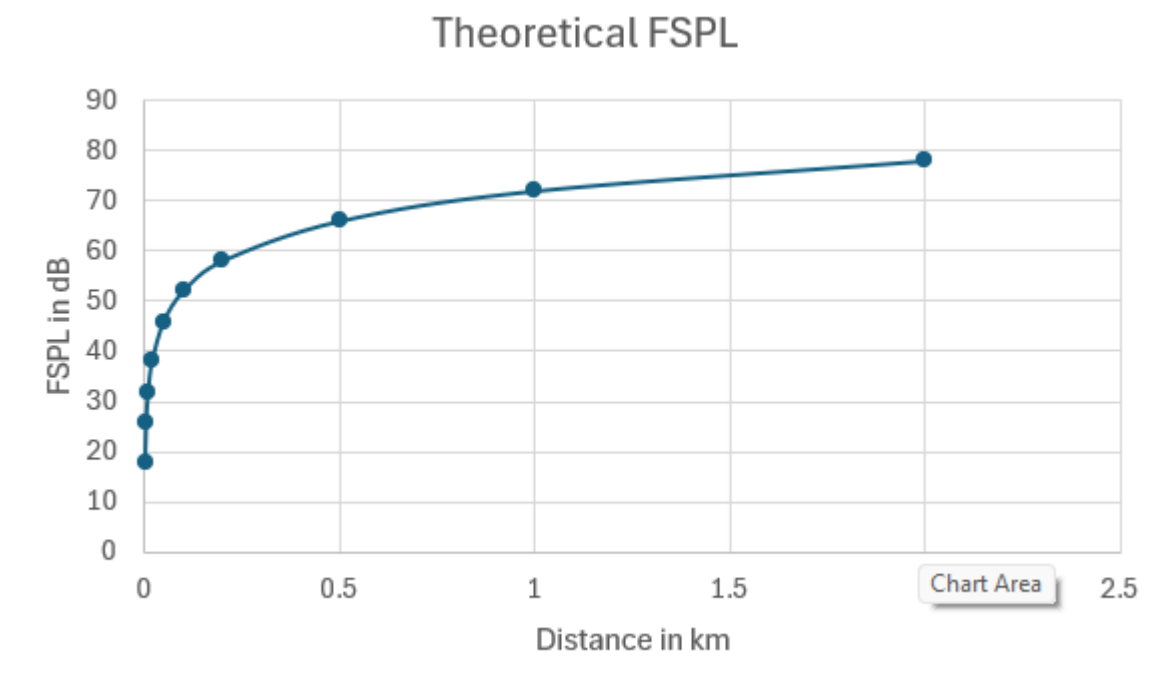

Figure *17: theoretical FSPL graph*

Conclusion

As shown in both graphs, it has been concluded that both methods deviate very little from each other. Both graphs follow the come pattern of rising exponentially and converging into a single value at the top. Therefore, we can conclude that exercise was a success as the simulated values do not diverge too much from theoretical values and can be used in the future for same environments. Therefore, it is implied the task could theoretically be applied for practical use without much error in accordance with this experiment.